网络安全新技术——

DeepLocker

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DeepLocker

- DeepLocker是什么？
- IBM 研究院在 2018 年的 Black Hat 大会上展示 的一种 AI 赋能的恶意代码 — DeepLocker，借助深度学习模型实现了对特定目标的精准定位与打击。

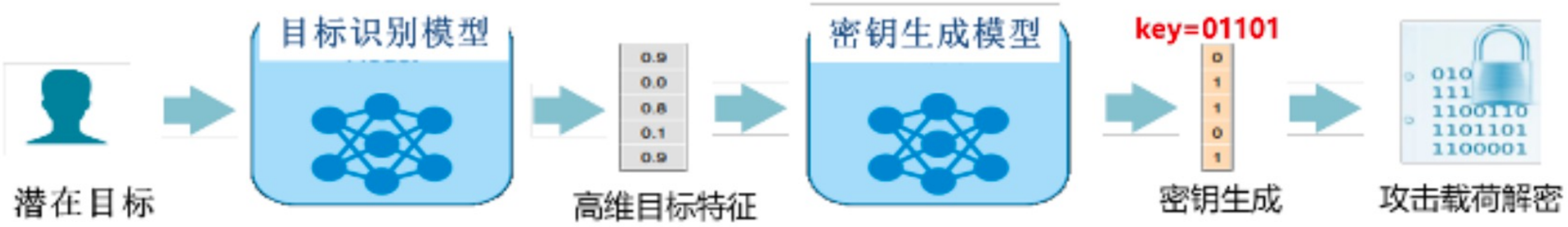


图 6 DeepLocker 攻击意图隐藏示意图

- 实现攻击意图隐藏的关键点就在于对密钥(用 于解密攻击载荷)的隐藏。
- DeepLocker 包含两个主要神经网络模型组件，即目标识别模型和密钥生成模型。

- 密钥的隐蔽存储与深度神经网络模型 实现的特征识别功能密切相关，目标特征未被检测 到时，神经网络模型便不会对正确的密钥做输出。

深度学习模型的“黑盒特性”，使得神经 网络模型难以被逆向分析

DeepLocker 利用了深度学习模型的泛化性，保证了生成密钥的稳定性

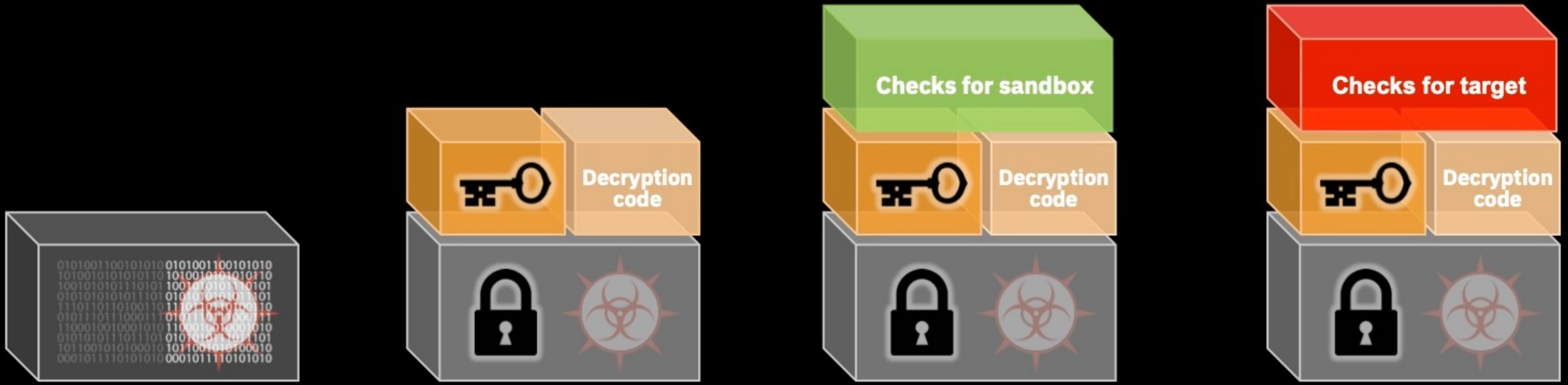
DeepLocker

Concealing Targeted Attacks with AI Locksmithing

Dhilung Kirat, Jiyong Jang, Marc Ph. Stoecklin

IBM **Research**

Malware concealment – Locksmithing



1980 — 1990 — 2000 — 2010 →

Obfuscation
Mutate payload

Polymorphism
Metamorphism

Encryption
Hide payload

Packers

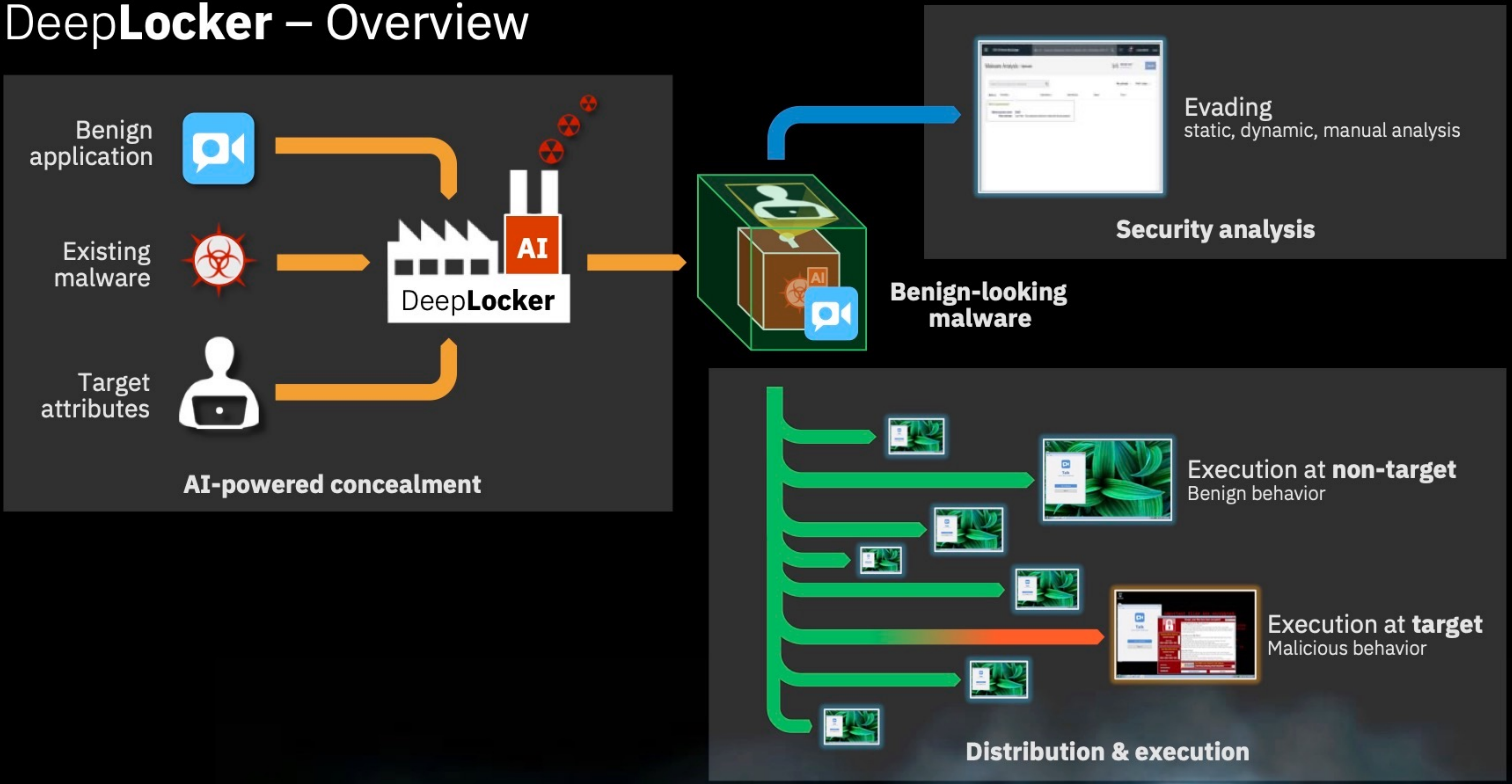
Evasive malware
Avoid being analyzed

VM check
Processes check

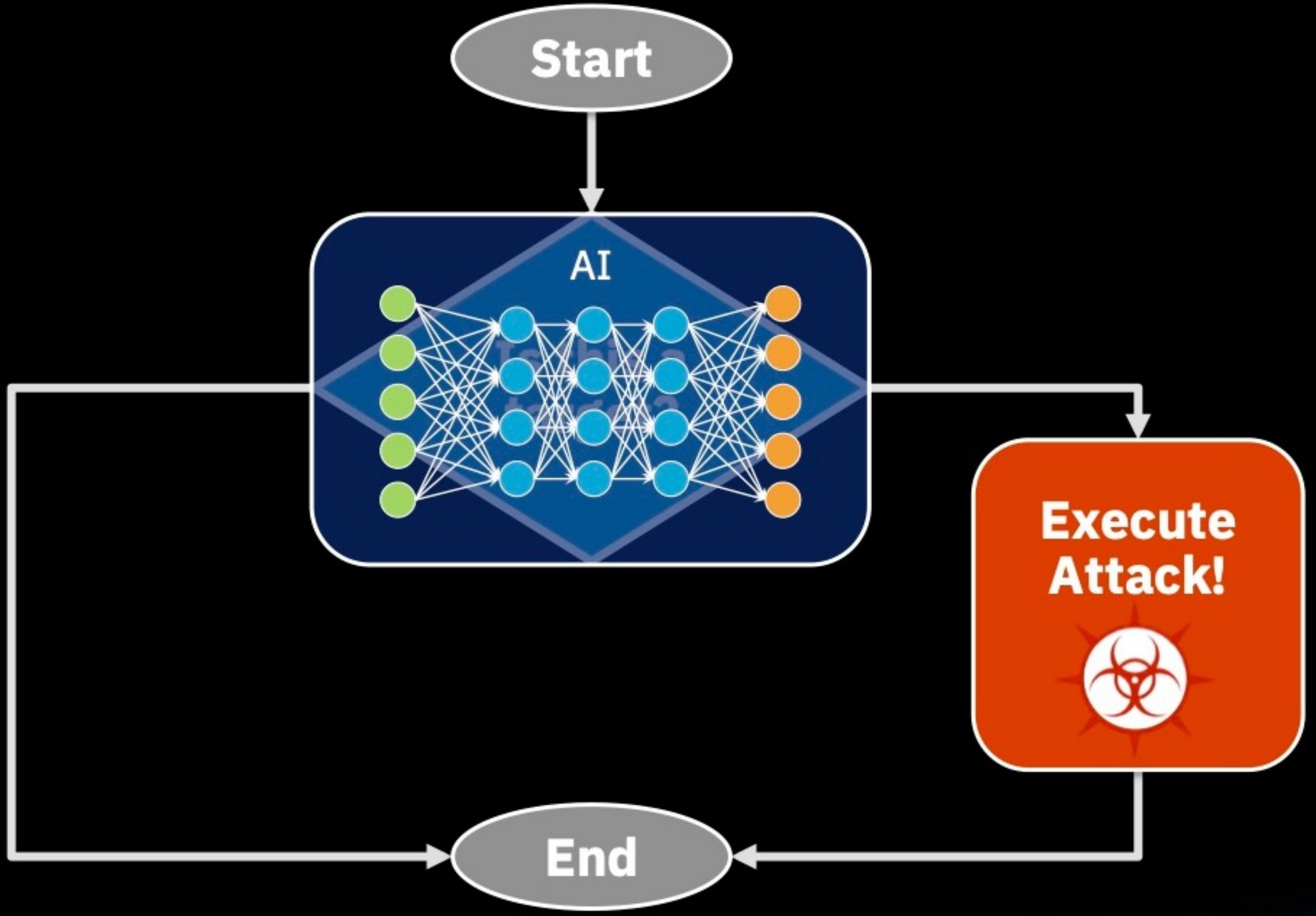
Targeted attack
Disclose only at a target

Target attributes check

DeepLocker – Overview

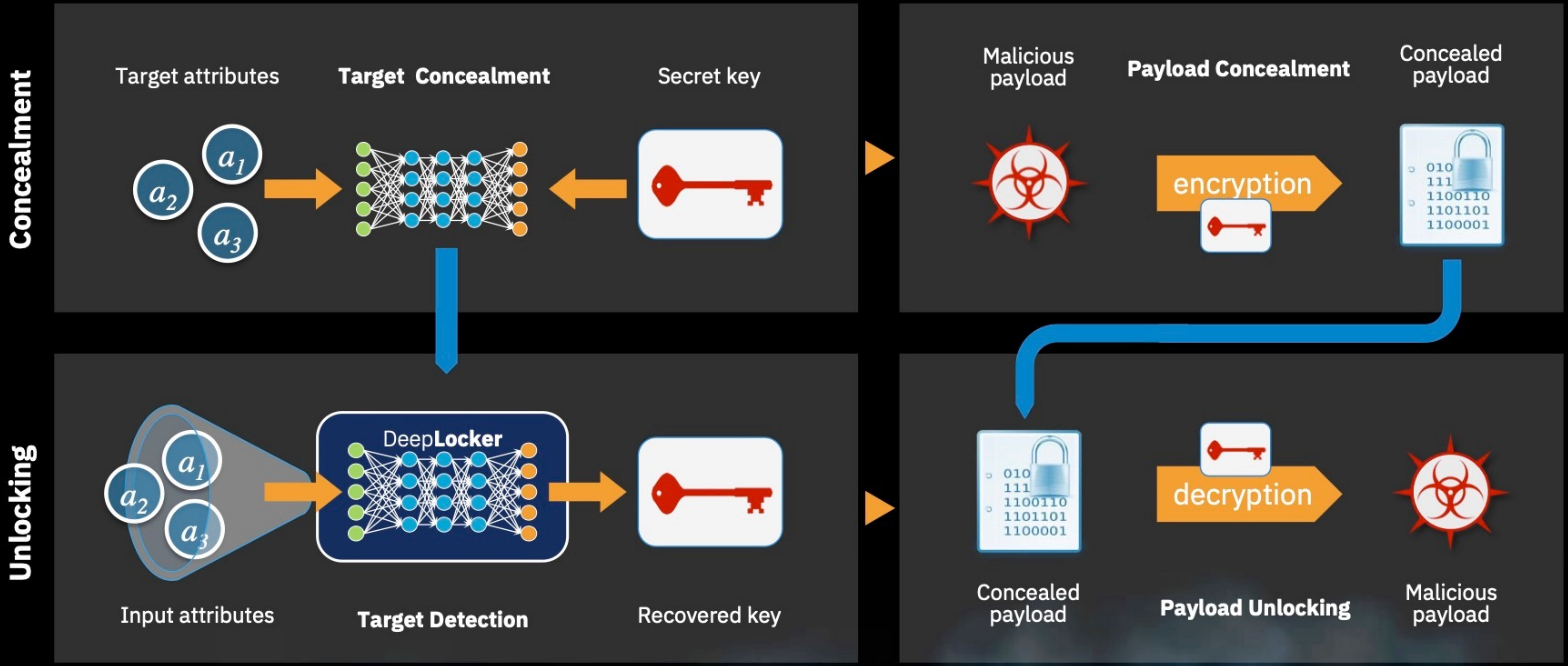


AI-powered targeted attack

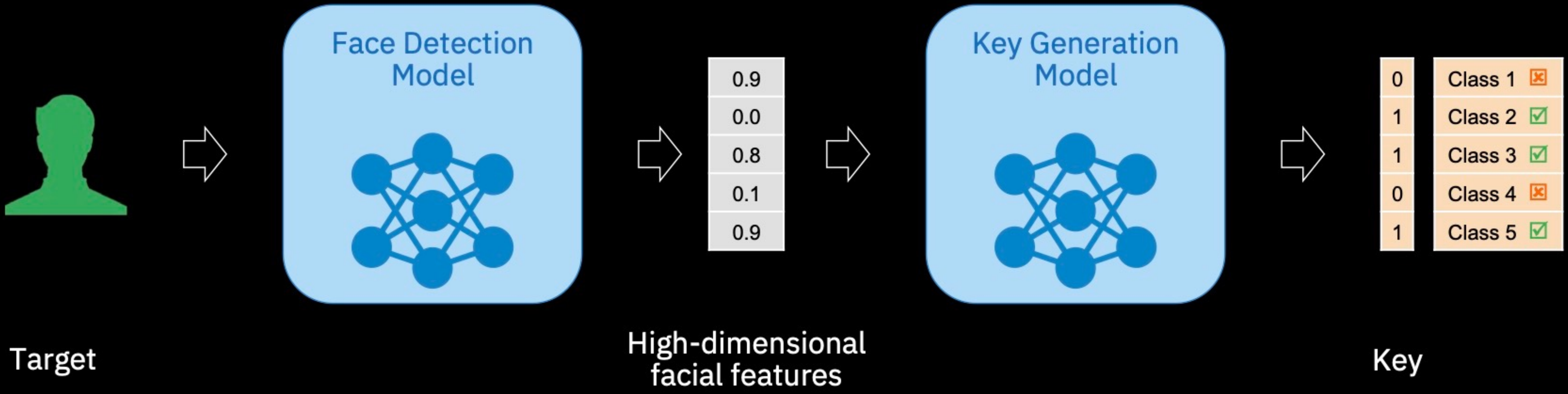


Target attributes

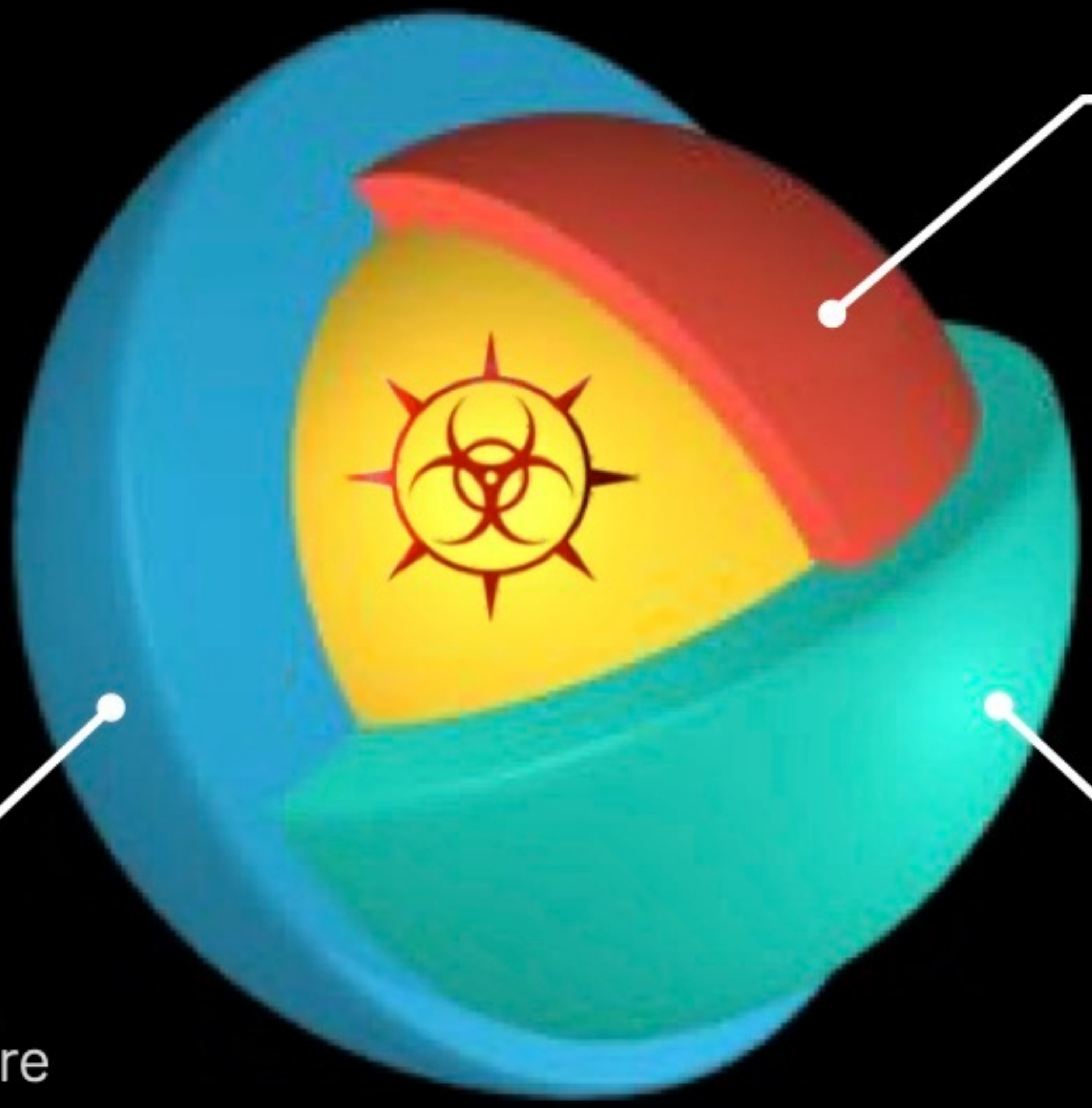
DeepLocker – AI-Powered Concealment and Unlocking



Key generation



Deep**Locker** – AI-powered concealment



3 Malicious Intent Concealment

Payload is fully encrypted concealing **how** the final attack is executed

1 Target Class Concealment

Does not reveal **what** it is looking for (e.g., faces, organization, or a completely obscure object specific to the target environment)

2 Target Instance Concealment

If the target class is an individual, it does not reveal **who** it is looking for

Thanks